The 1.5 × IQR Rule

Main idea: 1.5 is a widely used statistical convention for flagging potential outliers. It is not calculated from each dataset, and it is not a law of nature. The rule is useful for exploring data, but it does not prove that a flagged value is wrong.

1. What is an outlier?

An outlier is an observation that is unusually far from the rest of the data. It may be caused by a data-entry mistake, a measurement problem, a different population, or genuine but rare variation. A value should be investigated before it is removed or changed.

2. Understand Q1, Q3, and IQR first

Q1 (the first quartile) is the 25th percentile: roughly 25% of observations are at or below it. Q3 (the third quartile) is the 75th percentile: roughly 75% are at or below it. The interquartile range (IQR) measures the spread of the middle half of the data.

$$\mathbf{IQR = Q3 - Q1}$$

Quartile calculation methods can differ slightly between textbooks and software. For example, pandas uses a particular interpolation method by default. Use the method specified by your course or project when exact values matter.

3. The 1.5 × IQR rule and its formulas

First calculate IQR = Q3 − Q1. Then calculate the inner fences:

- **Lower fence = Q1 − 1.5 × IQR**
- **Upper fence = Q3 + 1.5 × IQR**

A value below the lower fence is flagged as a potential lower outlier. A value above the upper fence is flagged as a potential upper outlier. Values exactly on a fence are not beyond it, so the strict comparisons are < lower fence or > upper fence.

4. Why did they choose 1.5?

The 1.5 multiplier is associated with John Tukey's exploratory data analysis and boxplot approach. It became a practical convention for marking observations that are far from the central part of a dataset. It is a heuristic—a useful rule of thumb—not a value that can be uniquely derived for every dataset.

The goal is a balance: a fence that is too close to the quartiles can flag many ordinary observations, while a fence that is too far away may miss unusual values. The 1.5 multiplier became a widely adopted compromise for exploratory analysis.

A helpful normal-distribution explanation

If data follow a perfectly normal (bell-shaped) distribution, Q1 is approximately 0.674 standard deviations below the mean and Q3 is approximately 0.674 standard deviations above it. The normal-distribution IQR is therefore about 1.349 standard deviations. Multiplying that IQR by 1.5 gives about 2.023 standard deviations. The upper fence is then about 0.674 + 2.023 = 2.698 standard deviations above the mean, with a symmetric lower fence at about −2.698 standard deviations.

For ideal normal data, only about 0.7% of observations are expected to fall beyond these two fences combined (roughly 0.35% in each tail). This helps explain why the rule often flags relatively unusual values in bell-shaped data. This is an illustration, not proof that all datasets are normal or that 1.5 is universally optimal.

5. Why not use 1, 2, or 3?

Multiplier	Effect on fences	Typical interpretation
1.0 × IQR	Narrower fences	Flags more values; more sensitive
1.5 × IQR	Moderate fences	Common rule for potential outliers
3.0 × IQR	Wider fences	Flags only more extreme values

The choice changes sensitivity. A smaller multiplier usually flags more observations; a larger multiplier usually flags fewer. The right threshold depends on the purpose, distribution, and cost of false alarms versus missed unusual values.

6. Worked example

Suppose Q1 = 20 and Q3 = 40.

1. IQR = 40 − 20 = 20
2. 1.5 × IQR = 1.5 × 20 = 30
3. Lower fence = 20 − 30 = −10
4. Upper fence = 40 + 30 = 70

Example value	Result under the rule
−15	Potential lower outlier: below −10
50	Not flagged: between −10 and 70
80	Potential upper outlier: above 70

7. Inner fences versus outer fences

Some boxplots also use outer fences at 3 × IQR. These are intended to highlight more extreme observations:

- Lower outer fence = Q1 − 3 × IQR
- Upper outer fence = Q3 + 3 × IQR

The 1.5 × IQR fences flag potential outliers; the 3 × IQR fences flag more extreme potential outliers. Neither boundary alone determines whether a value is an error.

8. Important limitations

- The rule flags unusual values; it does not identify the reason they occurred.
- A genuine high salary, expensive house, or large transaction may be an outlier and still be valid.
- For skewed data, a one-sided distribution, or multiple groups mixed together, the rule may flag many values or miss context.
- Quartiles and fences can vary slightly depending on the quartile calculation method, especially in small datasets.
- Do not automatically delete or replace flagged observations. Check source records, context, and the goal of the analysis.

9. What should you do after finding an outlier?

- Check whether the value is a data-entry or measurement error.
- If it is wrong and the correct value is available, correct it from the trusted source.
- If it is invalid and cannot be corrected, consider excluding it with a documented reason.
- If it is genuine, consider keeping it, using a robust statistic/model, transforming the variable, or capping it only when justified.
- For machine learning, learn thresholds from the training data and apply the same thresholds to validation/test data to avoid data leakage.

10. Quick memory summary

- IQR = Q3 − Q1.
- Lower fence = Q1 − 1.5 × IQR.
- Upper fence = Q3 + 1.5 × IQR.
- 1.5 is a conventional multiplier associated with Tukey's boxplot method—not a value calculated from your dataset.
- The multiplier provides a practical balance for flagging unusual observations; it is not a universal law.
- A flagged value is a potential outlier, not automatically a mistake.

A final way to remember it

Think of 1.5 as a warning boundary, not a deletion command. It tells you, "This value is far enough from the middle 50% that it deserves a closer look." You then use subject knowledge and data-quality checks to decide what to do.